ThermoBench: Technical Submission Report

Candidate ID: RA2311026010405 | **Date:** November 2026 | **Project:** Windows 11 x64 Engineering Workbench (C++17)

■ **Google Drive Submission Package:**
https://drive.google.com/file/d/18Jt6KvRIcyc4wG4TsBWUNSlS6LfzkoJ6/...

■ **Video Walkthrough (3-5 min):**
https://drive.google.com/file/d/1oRRoOZ5wvf7-aGRu-NaxsGebJjBp9HAy/...

■ **Live Web App (Vercel):**
https://3d-varsha.vercel.app/

■ **GitHub Source Repository:**
https://github.com/var-13/3D

1. Executive Summary & Architecture Choices

ThermoBench is an offline-first Windows 11 engineering workbench uniting 1-D finite element thermal analysis, backpropagation surrogate training, multicore batch benchmarking, 3-D WebGL CAD/FEA inspection, and embedded telemetry simulation. The system enforces strict architectural separation:

- **Pure C++17 Numerics (src/core/)**: Decoupled core libraries with zero UI/graphics dependencies. Solves tridiagonal thermal systems in O(N) using the Thomas algorithm, verifies gradients analytically, and executes lock-free parallel batch runs.
- **Micro-Server Layer (src/server/)**: Lightweight embedded HTTP server (cpp-httplib) routing REST API requests and streaming static UI assets with zero external DLLs.
- **100% Offline Frontend (ui/)**: Single-page application utilizing vanilla JavaScript, Three.js r167 3-D viewport, and Chart.js 4.4.3 without any CDN or external network requirements.
- **Process Supervisor (Launch.exe)**: Win32 process manager verifying TCP health before auto-launching the default browser and guaranteeing clean child termination.

2. 1-D Thermal FEA Verification & Analytical Superconvergence

Governing Equation: **-d/dx (k A dT/dx) = q A** on x in [0, 1.0 m] with T(0)=100°C, T(L)=20°C, A=10■■ m², k=10 W/(m·K), q=10■ W/m³.
Exact Solution: **T(x) = T■ + ((T_L - T■)/L)x + (q / 2k) x (L - x)**. Solved using 2-node linear elements and the Thomas algorithm (TDMA):

Mesh (N)	Element h (m)	Midpoint Error (°C)	Theoretical 125·h² (°C)	Observed Ratio	Max Nodal Error
N = 10	0.1000	1.250000	1.250000	Baseline	< 1.0 × 10■¹¹ °C
N = 20	0.0500	0.312500	0.312500	4.0000 (Slope = 2.0)	< 1.0 × 10■¹¹ °C
N = 40	0.0250	0.078125	0.078125	4.0000 (Slope = 2.0)	< 1.0 × 10■¹¹ °C
N = 80	0.0125	0.019531	0.019531	4.0000 (Slope = 2.0)	< 1.0 × 10■¹¹ °C

Outcome: Midpoint error drops by exactly 4.0× per mesh doubling (O(h²) quadratic convergence, slope=2.0000). Nodal superconvergence is verified below 10■¹¹ °C.

3. Backpropagation Neural Network & Numerical Gradient Verification

A 3→4→1 feedforward neural network models nonlinear thermal relationships using tanh activations and MSE loss. Analytical chain-rule gradients ∂L/∂θ were cross-verified against central finite differences (ε=10■■):
max |∂L/∂θ_analytic - ∂L/∂θ_FD| < 4.2 × 10■■ across all weights and biases. A 100-epoch training run validated monotonic convergence from L■=0.4218 to L■■■=0.0142 (saved in results/backprop_loss.csv).

4. Multicore Batch Parallelism & Scaling Benchmark

Batch thermal analysis processes 200 deterministic test cases distributed dynamically via lock-free atomic counters (std::atomic<size_t>). Numerical parity across all cases satisfies: **max |T_parallel - T_serial| = 0.00000000 °C**.

Thread Count	Median Runtime (5 Runs)	Speedup S = T■ / T■	Parallel Efficiency	Numerical Parity Check
1 Thread (Serial)	18.42 ms	1.00×	100.0%	Reference (0.00000000 °C)
2 Threads	9.68 ms	1.90×	95.1%	Exact Match (< 10■¹² °C)
4 Threads	5.21 ms	3.54×	88.4%	Exact Match (< 10■¹² °C)
8 Threads	3.12 ms	5.90×	73.8%	Exact Match (< 10■¹² °C)

5. Embedded Telemetry Subsystem & Fault-Tolerant Design

The embedded module (src/core/embedded_mock.cpp) models three concurrent subsystem target devices:

- **MCU-STM32H743**: Core controller monitoring supply rails (3.3V) and board junction temperatures.
- **FPGA-ARTIX7**: Hardware acceleration matrix engine operating on 1.0V internal logic.
- **JTAG-FT232H**: High-throughput diagnostic and telemetry bridge transport.

Fault Injection & Autonomous Recovery: Watchdog timeout faults simulate bus stall conditions, transitioning the device to a FAULTED state with status alerts. The built-in soft-reset routine re-initializes registers, re-enumerates the peripheral bus, and restores OPERATIONAL status without requiring a server reboot.

6. 3-D Geometry, CAD & ANSYS Parsers

- **ASCII STL Parser**: Validates vertex finiteness, normal vectors, and bounding box. Accurately parses 12-facet heatsink CAD models and computes total surface area (0.0070 m²).
- **ANSYS CSV Point Cloud Parser**: Ingests 65 thermal FEA nodal coordinates and temperatures (28.2 °C to 94.8 °C), rendering an interactive Three.js 3-D thermal colormap with point inspection.

7. Test Suite Outcomes & Verification Evidence

```
Total Test Suite: 61 / 61 Unit Tests PASSED (0 Failures, 0 Compiler Warnings)
• Phase 0: Build & Launcher Setup ..... [ 3 / 3 PASSED ]
OK
• Phase 1: 1-D Thermal FEA Solver (Thomas TDMA, Superconvergence) ..... [ 22 / 22 PASSED ]   OK
• Phase 2: Backpropagation Engine (Finite Difference Check) ..... [ 10 / 10 PASSED ]   OK
• Phase 3: Multicore Batch Solver (Atomic Scheduling, Parity) ..... [ 12 / 12 PASSED ]   OK
• Phase 4: STL & ANSYS Geometry Parsers ..... [ 8 / 8 PASSED ]   OK
• Phase 5: Embedded Mock & Watchdog Telemetry ..... [ 6 / 6 PASSED ]   OK
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8. Known Limitations

1. **Binary STL Support**: Currently accepts standard ASCII STL files; binary STL format requires future binary parser extension.
2. **Transport Security**: The local daemon serves standard HTTP for low-overhead local IPC. Production deployments should sit behind a TLS reverse proxy.
3. **Synthetic Telemetry**: Hardware metrics are simulated algorithmically; physical hardware requires physical SPI/I2C/CAN transceivers.

9. AI-Assisted Development Disclosure & Ownership Statement

- **Original Implementation**: All numerical algorithms (FEM, Thomas solver, gradient derivations, batch solver, geometry parsers, embedded mock), launcher, UI layout, test cases, and packaging scripts were authored specifically for this project.
- **Permissive Open-Source Libraries Reused**: cpp-httplib (v0.16.3, MIT), Chart.js (v4.4.3, MIT), Three.js (r167, MIT).
- **AI Log Reference**: AI interactions (Claude Sonnet / Gemini Flash) are cataloged in docs/AI_LOG.md, including genuine corrections made during toolchain configuration and process handle lifecycle management.

10. Complete Submission Package & Access Links

All code, build scripts, raw CSV results, labelled plots, documentation, and version-control bundles are hosted at:

- **Google Drive Submission Package:** https://drive.google.com/file/d/18Jt6KvRIcyc4wG4TsBWUNSlS6LfzkoJ6/view?usp=sharing
- **Demonstration Video (3–5 min):** https://drive.google.com/file/d/1oRRoOZ5wvf7-aGRu-NaxsGebJjBp9HAy/view?usp=sharing
- **Live Interactive Web Application:** https://3d-varsha.vercel.app/
- **GitHub Source Code Repository:** https://github.com/var-13/3D